

Problem

After each operation, print the number of set bits in the respective BitSets as two space-separated integers on a new line.

Input Format

The first line contains **2** space-separated integers, **N** (the length of both BitSets **B_1** and **B_2**) and **M** (the number of operations to perform), respectively.

Submissions

The **M** subsequent lines each contain an operation in one of the following forms:

- **AND** **<set>** **<set>**
- **OR** **<set>** **<set>**
- **XOR** **<set>** **<set>**
- **FLIP** **<set>** **<index>**
- **SET** **<set>** **<index>**

Leaderboard

In the list above, **<set>** is the integer **1** or **2**, where **1** denotes **B_1** and **2** denotes **B_2** .

<index> is an integer denoting a bit's index in the BitSet corresponding to **<set>**.

Discussions

For the binary operations **AND**, **OR**, and **XOR**, operands are read from left to right and the BitSet resulting from the operation replaces the contents of the first operand. For example:

AND 2 1

B_2 is the left operand, and **B_1** is the right operand. This operation should assign the result of **$B_2 \wedge B_1$** to **B_2** .

Editorial

Constraints

- $1 \leq N \leq 1000$
- $1 \leq M \leq 10000$

Output Format

After each operation, print the respective number of set bits in BitSet **B_1** and BitSet **B_2** as **2** space-separated integers on a new line.

Sample Input

```
5 4
AND 1 2
SET 1 4
FLIP 2 2
OR 2 1
```

Sample Output

```
0 0
```

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Language

Java 15



```
1 import java.io.*;
2 import java.util.*;
3
4 public class Solution {
5
6     public static void main(String[] args) {
7         /* Enter your code here. Read input from STD
8            output to STDOUT. Your class should be namec
9     }
```

Line: 9 Col: 2

Upload Code as File

Run Code

Submit Code



Test against custom input

